

ECO101 - Introduction to Microeconomics

Department of Economics and Social Science (ESS), Brac University

Government interventions: Welfare and efficiency

Lecture: 4

Reading

- Economics by Michael Parkin, 10th edition
 - Chapter: 5 & 6
- Principles of Economics by Mankiw, 6th Edition
 - Chapter: 6, 7 & 8

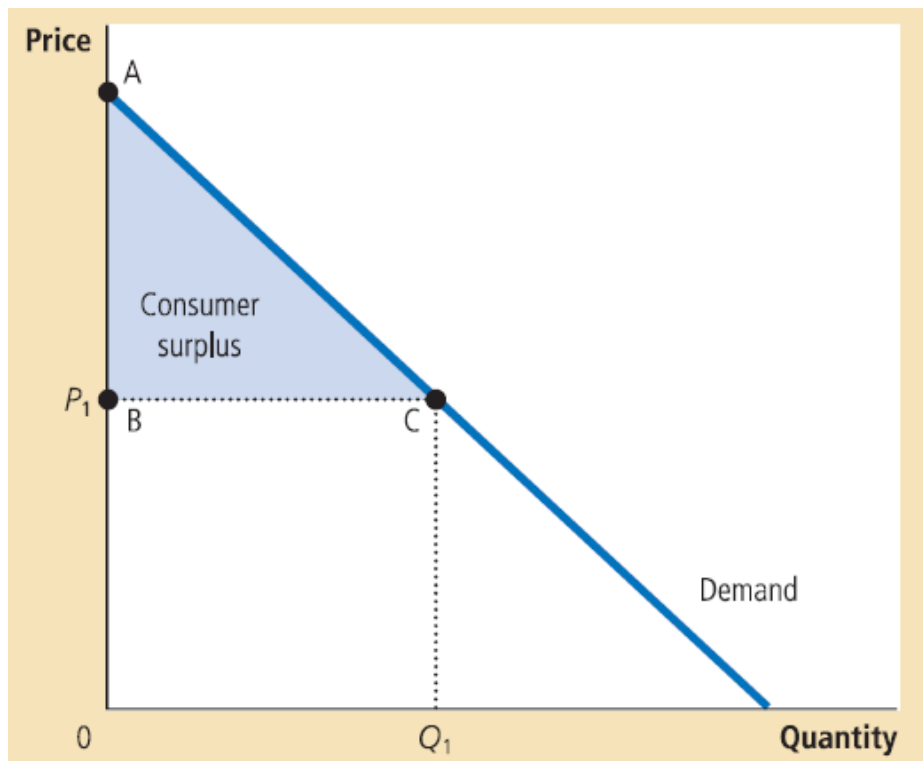
Introduction

- Is there a “right price for a good” from the standpoint of society as a whole?
- To answer this question, we take up welfare economics, the study of how the allocation of resources affects economic well-being
- This analysis leads to a profound conclusion: The equilibrium of supply and demand in a market maximizes the total benefits received by buyers and sellers

Demand, Willingness to Pay & Value

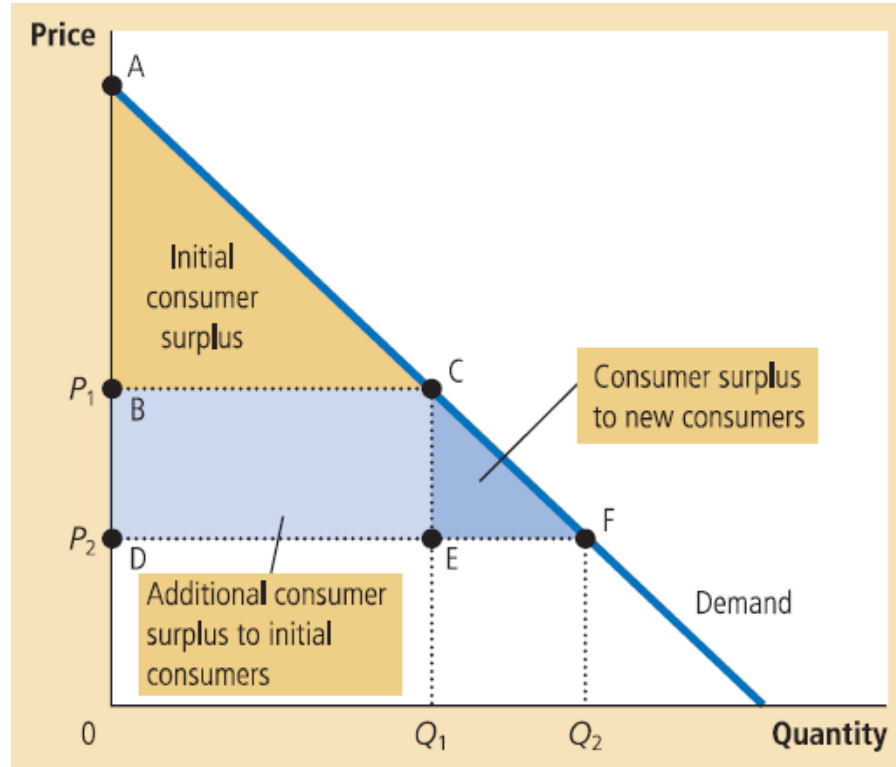
- We commonly use the expression “getting value for money”, therefore, distinguishing between value and price
- The value of one more unit of a good or service is its marginal benefit
- Marginal benefit is measured by the maximum price that a consumer is willing to pay for another unit of the good or service
- Willingness to pay determines demand  A demand curve is a marginal benefit curve

Consumer Surplus

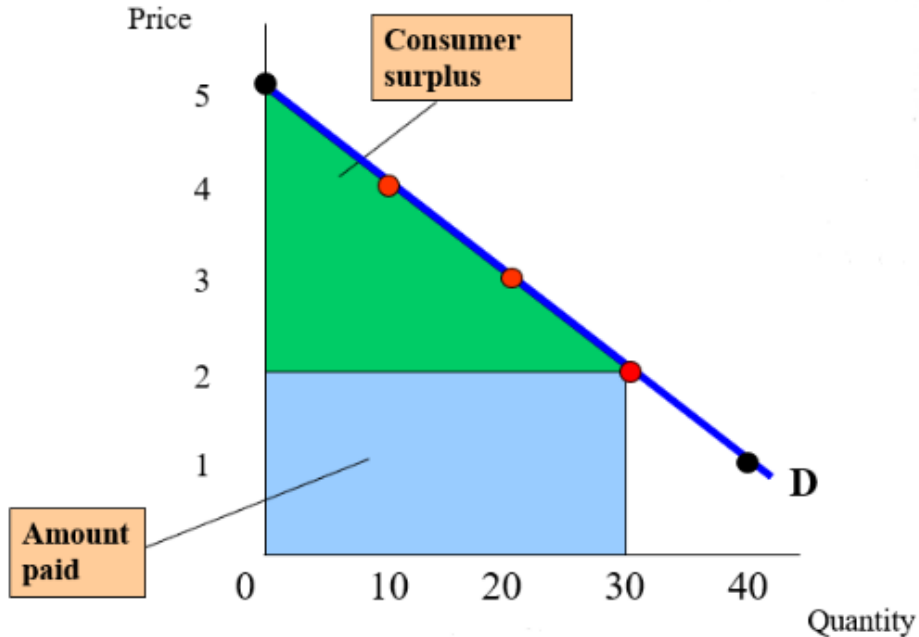


- **Consumer surplus:** Consumer surplus is the excess of the benefit received from a good over the amount paid for it

Consumer Surplus - Graphical Representation



Measuring Consumer Surplus



The consumer surplus is simply the area of the green triangle

Area of a triangle: $\frac{1}{2} \times \text{base} \times \text{height}$

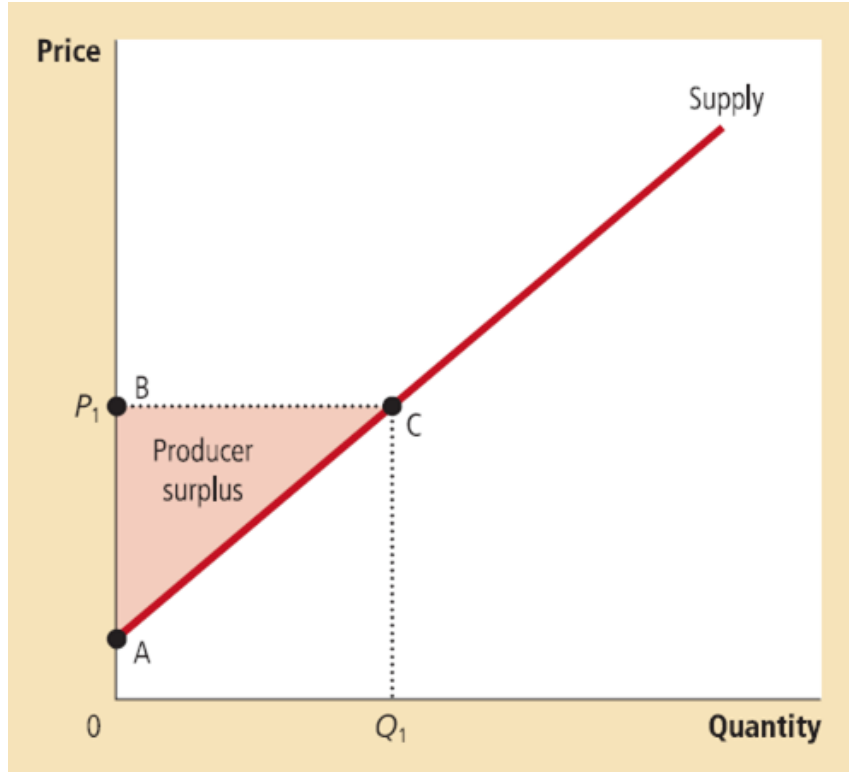
$$CS = \frac{1}{2} \times 30 \times (5-2) = 45$$

Consumer surplus measures consumer welfare in a market

Producer Surplus

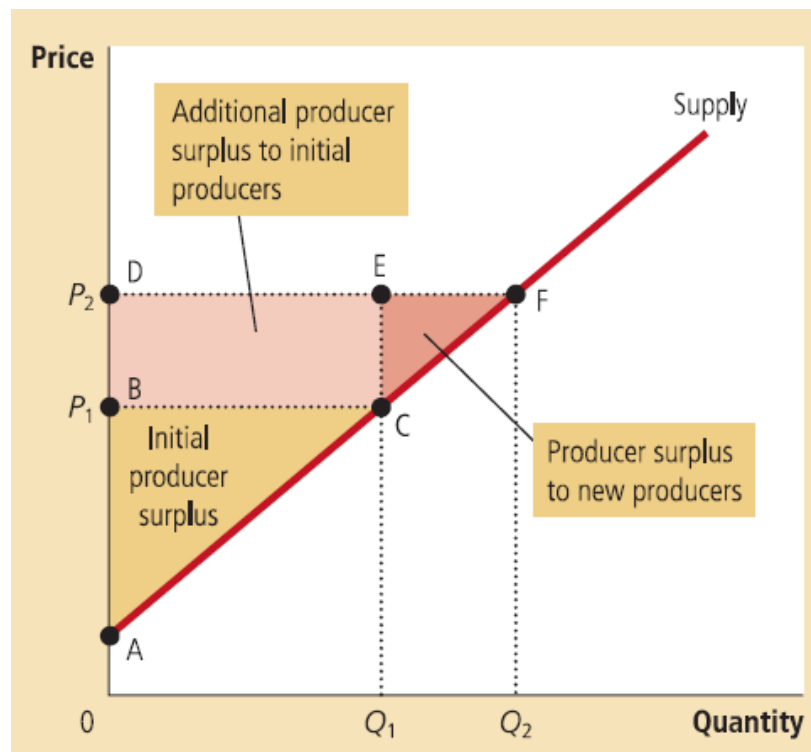
- Just as consumers distinguish between value and price, so do producers distinguish between cost and price
- The cost of producing one more unit of a good or service is its marginal cost
- Marginal cost is the minimum price that producers must receive to induce them to offer one more unit of a good or service for sale
- A supply curve is a marginal cost curve

Producer Surplus

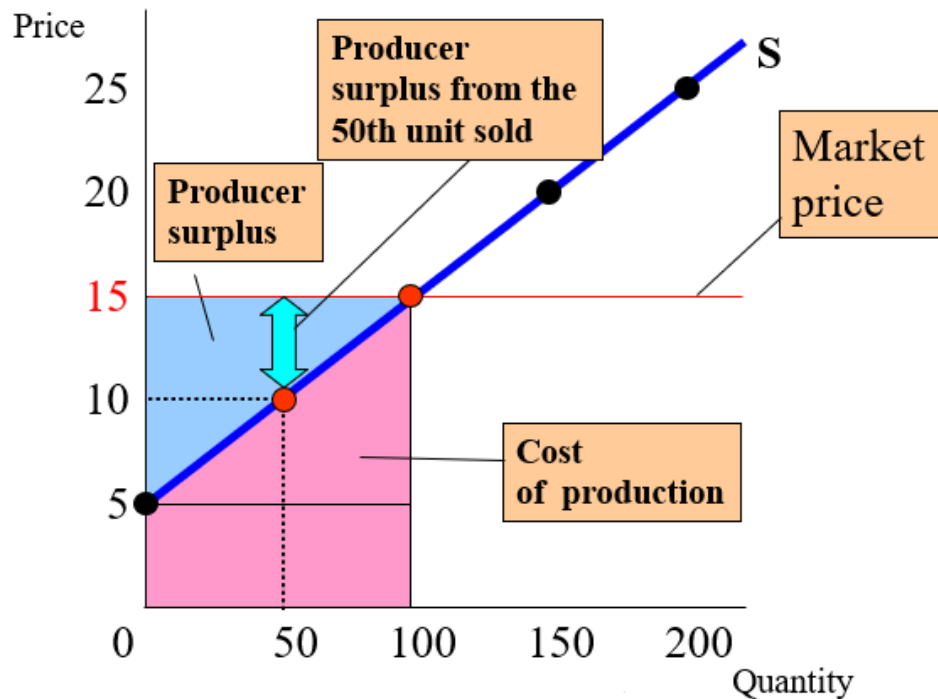


- **Producer surplus:** Producer surplus is the excess of the amount received from the sale of a good or service over the cost of producing it

Producer Surplus - Graphical Representation

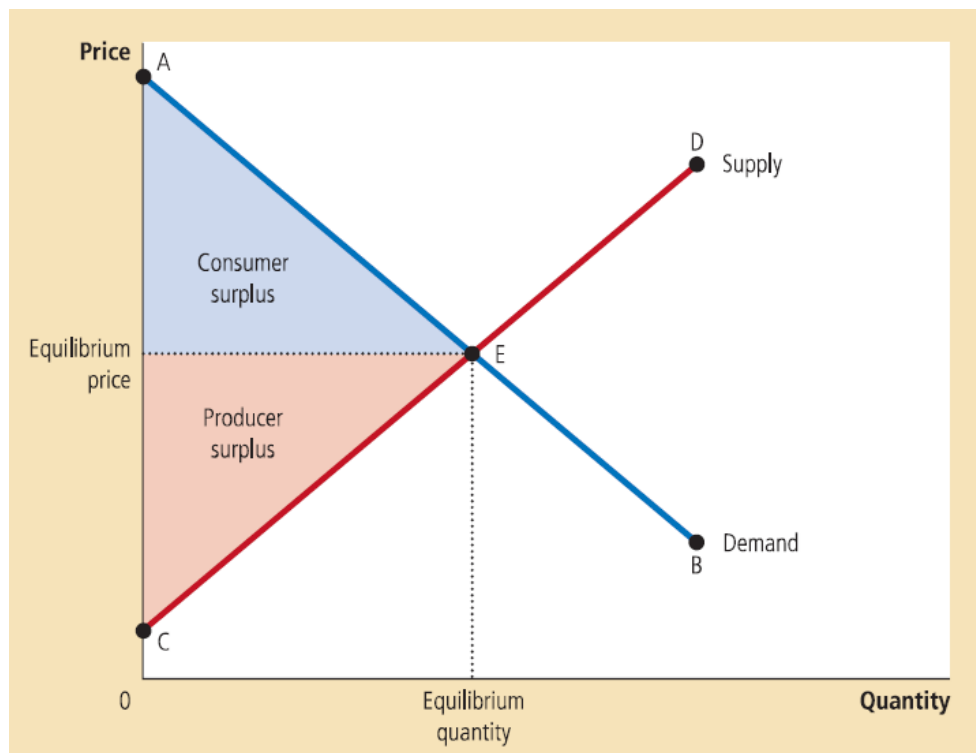


Measuring Producer Surplus



- The producer surplus is simply the area of the blue triangle
- Area of a triangle:
 $\frac{1}{2} \times \text{base} \times \text{height}$
- $PS = \frac{1}{2} \times 100 \times (15 - 5) = 500$

Consumer & Producer Surplus in Market Equilibrium



- The sum of Consumer and Producer surplus is called **Total Surplus**, which is the area between the demand and supply curves up to the equilibrium quantity

Price Controls

- Price controls are usually enacted when policymakers believe that the market price of a good or service is unfair to buyers or sellers
- Yet, as we will see, these policies can generate inequities of their own
- Price ceiling a legal maximum on the price at which a good can be sold
- Price floor a legal minimum on the price at which a good can be sold

Price Ceiling - CS, PS & DWL



Price Ceiling - Numerical Application

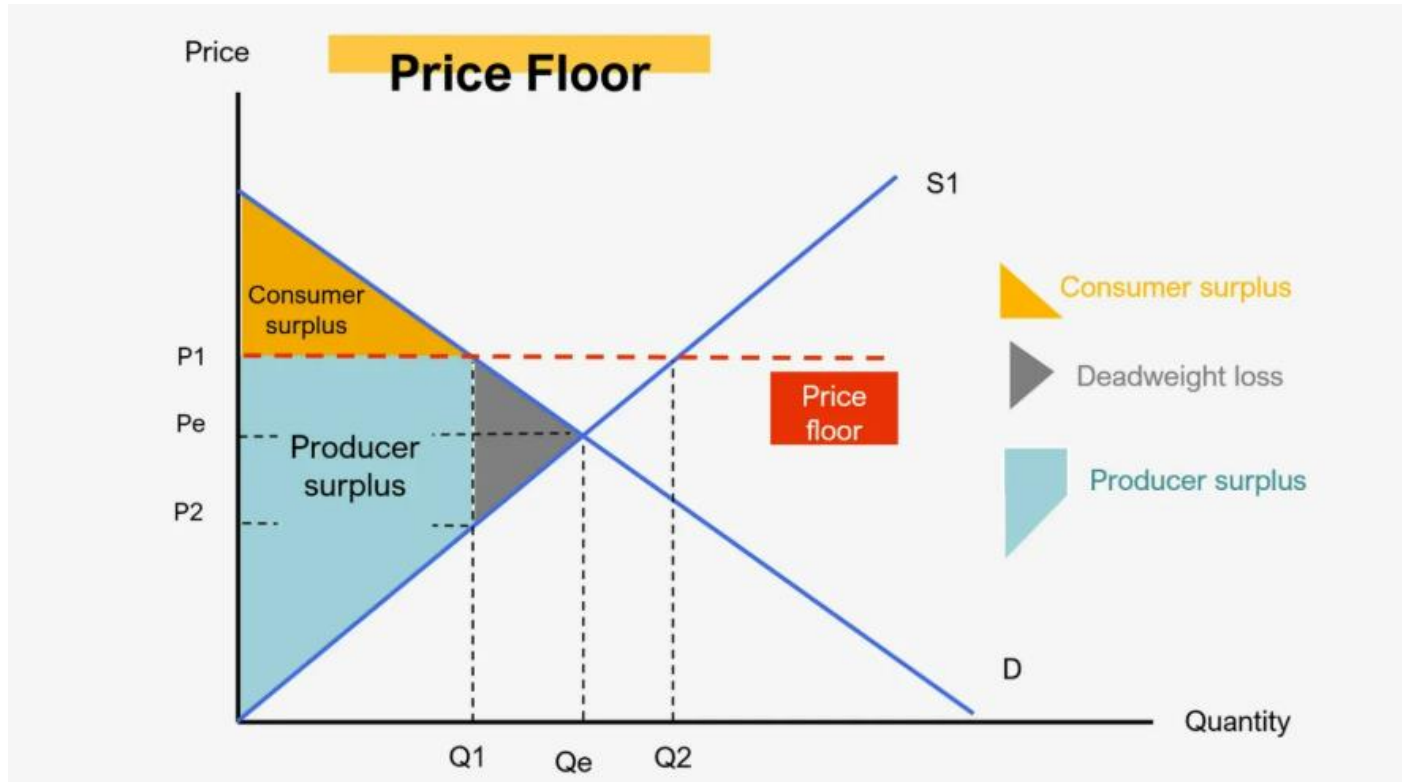
Suppose that the market for gasoline can be represented by the following equations:

$$\text{Demand: } P = 10 - 2Q_D$$

$$\text{Supply: } P = 1 + 0.5Q_S$$

- a) Calculate the equilibrium price and quantity of gasoline.
- b) Calculate the consumer surplus and producer surplus at the initial equilibrium price and quantity.
- c) If the government sets a price ceiling of \$2.25 per gallon of gasoline, what is new consumer surplus and producer surplus with the price ceiling of \$2.25 per gallon?
- d) Calculate the deadweight loss.

Price Floor - CS, PS & DWL



Price Floor - Numerical Application

Suppose that the market for milk can be represented by the following equations:

$$\text{Demand: } P = 12 - 0.5Q_D$$

$$\text{Supply: } P = 0.1Q_S$$

- a) Calculate the equilibrium price and quantity of milk.
- b) To help dairy farmers, the government sets a minimum price of \$2.50 per gallon of milk. What is the new quantity of milk sold in the marketplace? Illustrate your answers to (a) and (b) on a graph.
- c) Using this graph, calculate the consumer surplus and producer surplus before and after the price floor is imposed.
- d) Calculate the deadweight loss.

Market Failure and Deadweight Loss

- Markets do not always achieve an efficient outcome
- We call a situation in which a market delivers an inefficient outcome is one of the market failures (inability of some unregulated markets to allocate resources efficiently)
- Market failure can occur because too little of an item is produced (underproduction) or too much is produced (overproduction)
- Therefore, due to under or overproduction caused by market failure, the quantity produced is inefficient
- And we measure the scale of inefficiency by **deadweight loss**, which is the decrease in total surplus that results from inefficient level of production

Market Failure and Deadweight Loss

Inefficient production creates a deadweight loss that is borne by the entire society: it is a social loss

Obstacles to efficiency that bring market failure and create deadweight losses are

- Price and quantity regulations
- **Taxes** and subsidies
- Externalities
- Public goods and common resources
- Monopoly
- High transaction costs

For ECO101 we will focus on Taxes

A Tax on Sellers

Tax Incidence

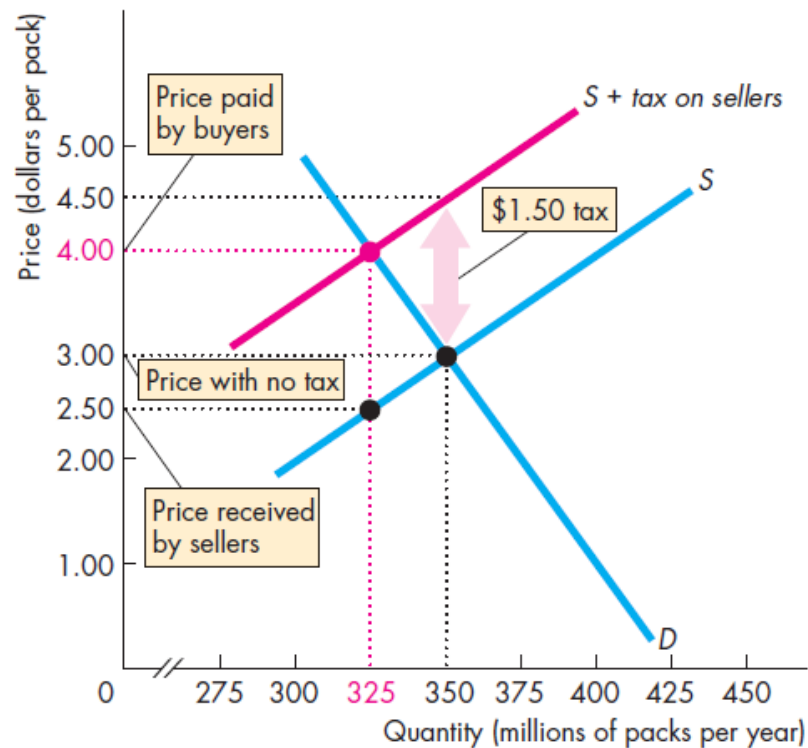
- Taxes increase the prices paid by buyers and lower the prices received by sellers
- Tax incidence is the division of the burden of a tax between buyers and sellers
- When the government imposes a tax on the sale of a good, the price paid by buyers might rise by the full amount of the tax, by a lesser amount, or not at all
- If the price paid by buyers rises by a lesser amount than the tax, then the burden on the tax falls partly on buyers and partly on sellers
- And if the price paid by buyers doesn't change at all, then the burden of the tax falls entirely on sellers

Tax on Sellers

Suppose On July 1, 2002, Mayor Bloomberg put a tax of \$1.50 a pack on cigarettes sold in New York City

To work out the effects of this tax on the sellers of cigarettes, we begin by examining the effects on demand and supply in the market for cigarettes

Tax on Sellers - Example



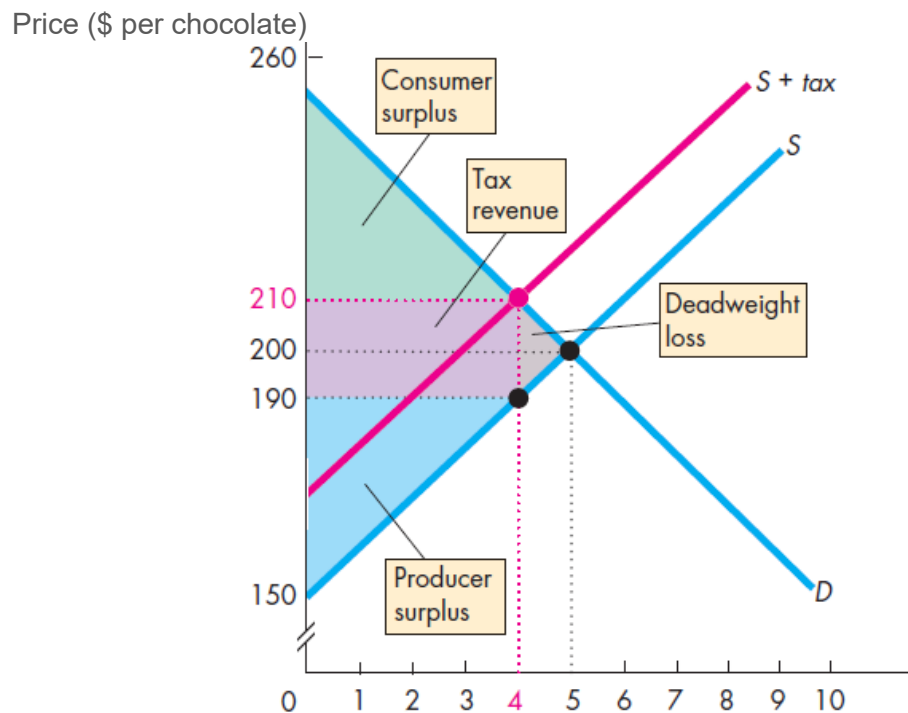
Taxes and Efficiency

- A tax drives a wedge between the buying price and the selling price and results in inefficient underproduction
- The price buyers pay is also the buyers' willingness to pay, which measures marginal social benefit
- The price sellers receive is also the sellers' minimum supply-price, which equals marginal social cost
- A tax makes marginal social benefit exceed marginal social cost, shrinks the producer surplus and consumer surplus, and creates a deadweight loss

Taxes and Efficiency - Example

- Suppose, in a market for chocolates 5000 quantity of chocolates (that is efficient quantity) are produced at price \$200 with no tax
- With a tax of \$20, the sellers' minimum supply-price rises by the amount of the tax and the supply curve shifts to $S + \text{tax}$
- The buyer's price rises to \$210, the sellers' price falls to \$190, and the quantity decreases to 4,000 chocolates a week
- As a result, both consumer surplus and the producer surplus shrinks
- Part of the loss of consumer surplus and producer surplus goes to the government as tax revenue (the purple area) and part becomes a deadweight loss (the gray area)

Taxes and Efficiency - Example



Tax revenue that government collects equals $T \times Q$ (the size of the tax times the quantity sold)

Tax on Seller

Consider the market for lollipops

Market Demand: $P = 220 - \frac{3}{5}Q$

Market Supply: $P = 30 + \frac{2}{5}Q$

- Find the market equilibrium price and quantity for lollipops.
- Assume that the government imposes a tax of \$50 on lollipops which must be paid by sellers. What is the new equilibrium price and quantity? Illustrate in a well labelled diagram the after effects of tax in the market for lollipops.
- Calculate the consumer and producer surplus.
- Calculate the deadweight loss and government tax revenue resulting from imposition of tax.

Graphical and mathematical/numerical analysis for all topics —
Price Ceiling, Price Floor and Tax on Seller demonstrated
during the lecture — are all **examinable**